Unit 8.01: PCA & Dimensionality Reduction

Q1. Principal Component Analysis PCA projects a dataset from mathbb{R}^{50} onto its top 6 principal components. What is the dimension of each transformed data point?

[ ] A: 25 [ ] B: 50
[ ] C: 44 **[x] D: 6**

Solution: PCA projection onto k eigenvectors transforms each sample vector from mathbb{R}^D to mathbb{R}^k . Here k = 6 , so the reduced dimension is 6 .

Q2. A 3D dataset has covariance matrix eigenvalues lambda_1 = 62 , lambda_2 = 24 , and lambda_3 = 14 . What percentage of total variance is explained by the first principal component PC_1 ?

[x] A: 62.0% [ ] B: 86.0%
[ ] C: 24.0% [ ] D: 72.1%

Solution: Total variance is sum lambda_i = 62 + 24 + 14 = 100 . The proportion explained by PC_1 is frac{lambda_1}{sum lambda_i} = frac{62}{100} = 62.0% .

Q3. A 3D dataset has covariance matrix eigenvalues lambda_1 = 64 , lambda_2 = 23 , and lambda_3 = 13 . What percentage of total variance is explained by the first principal component PC_1 ?

[x] A: 64.0% [ ] B: 87.0%
[ ] C: 23.0% [ ] D: 73.6%

Solution: Total variance is sum lambda_i = 64 + 23 + 13 = 100 . The proportion explained by PC_1 is frac{lambda_1}{sum lambda_i} = frac{64}{100} = 64.0% .

Q4. A 3D dataset has covariance matrix eigenvalues lambda_1 = 66 , lambda_2 = 22 , and lambda_3 = 12 . What percentage of total variance is explained by the first principal component PC_1 ?

[x] A: 66.0% [ ] B: 88.0%
[ ] C: 22.0% [ ] D: 75.0%

Solution: Total variance is sum lambda_i = 66 + 22 + 12 = 100 . The proportion explained by PC_1 is frac{lambda_1}{sum lambda_i} = frac{66}{100} = 66.0% .

Q5. A 3D dataset has covariance matrix eigenvalues lambda_1 = 68 , lambda_2 = 21 , and lambda_3 = 11 . What percentage of total variance is explained by the first principal component PC_1 ?

[x] A: 68.0% [ ] B: 89.0%
[ ] C: 21.0% [ ] D: 76.4%

Solution: Total variance is sum lambda_i = 68 + 21 + 11 = 100 . The proportion explained by PC_1 is frac{lambda_1}{sum lambda_i} = frac{68}{100} = 68.0% .

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Q6. A 3D dataset has covariance matrix eigenvalues lambda_1 = 70 , lambda_2 = 20 , and lambda_3 = 10 . What percentage of total variance is explained by the first principal component PC_1 ?

[x] A: 70.0%
[ ] B: 90.0%
[ ] C: 20.0%
[ ] D: 77.8%

Solution: Total variance is sum lambda_i = 70 + 20 + 10 = 100 . The proportion explained by PC_1 is frac{lambda_1}{sum lambda_i} = frac{70}{100} = 70.0% .

Q7. A 3D dataset has covariance matrix eigenvalues lambda_1 = 72 , lambda_2 = 19 , and lambda_3 = 9 . What percentage of total variance is explained by the first principal component PC_1 ?

[x] A: 72.0%
[ ] B: 91.0%
[ ] C: 19.0%
[ ] D: 79.1%

Solution: Total variance is sum lambda_i = 72 + 19 + 9 = 100 . The proportion explained by PC_1 is frac{lambda_1}{sum lambda_i} = frac{72}{100} = 72.0% .

Q8. A 3D dataset has covariance matrix eigenvalues lambda_1 = 74 , lambda_2 = 18 , and lambda_3 = 8 . What percentage of total variance is explained by the first principal component PC_1 ?

[x] A: 74.0%
[ ] B: 92.0%
[ ] C: 18.0%
[ ] D: 80.4%

Solution: Total variance is sum lambda_i = 74 + 18 + 8 = 100 . The proportion explained by PC_1 is frac{lambda_1}{sum lambda_i} = frac{74}{100} = 74.0% .